Mathematical Explanation of the Bayesian 2AFC Models

Models M0 through M4 | sensory inference, hidden-block belief, learning, and duration-aware switching

Core idea

All five models use the same noisy sensory measurement and Bayesian choice mechanism. They differ primarily in how they construct the trialwise prior probability that the current stimulus category is long.

Model	Prior mechanism	New fitted component	Parameters
M0	Fixed prior π=0.5	Sensory choice only	5
M1	Constant-hazard HMM	Hazard with fixed block prototypes	6
M2	Constant-hazard HMM	Fitted static subjective prototypes	7
M3	HMM plus learning	Feedback-driven prototype contrast	8
M4	Duration-aware HSMM	Run-length-dependent hazard slope	9

Scope

This document explains the equations implemented in the repository. It separates the sensory readout, the latent-state recursion, the fitting objective, exact model restrictions, and interpretation cautions. It does not alter the model code.

1. Notation and model architecture

For session d and trial t:

Symbol	Meaning
s_{dt}	Physical stimulus duration in seconds.
X_{dt}	Noisy internal measurement of duration.
$C_{dt} \in \{0,1\}$	Presented short/long category.
$Y_{dt} \in \{0,1\}$	Observed short/long choice.
π_{dt}	Pre-choice prior probability that the category is long.
q_{dt}	Long-choice probability before lapse mixing.
p_{dt}	Final observable long-choice probability.
Φ	Standard-normal cumulative distribution function.
$Z_t \in \{N,S,L\}$	Latent neutral, short-biased, or long-biased block.

The complete computation can be summarized as:

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past feedback -> latent belief -> π -> Bayesian sensory decision -> q
current physical category + lapse rates -> p -> choice likelihood
current feedback -> posterior belief -> learning -> next-trial transition
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The dynamic models are causal in the sense that current category feedback is applied only after forming the current Bayesian prior. A qualification is discussed in Section 4: the current category still selects the current lapse rate.

2. Shared sensory measurement model

The animal does not act directly on the physical duration. It acts on a noisy internal measurement:

$$X \mid s \sim \text{Normal}(s, \sigma^2)$$
$$\sigma^2 = w\, s^2 + s_0^2$$

- s_0 is baseline sensory standard deviation, in seconds.
- w controls duration-dependent variance. When baseline noise is small, the standard deviation is approximately $\sqrt{w} \cdot s$, so $\sqrt{w}$ is the Weber-like scale.

The two stimulus categories have fixed empirical duration moments:

$$\mu_k = (1/N_k) \sum_i s_i$$
$$\tau_k^2 = (1/N_k) \sum_i (s_i - \mu_k)^2,\ k \in \{0,1\}$$

The distribution of internal measurements associated with category k is approximated as

$$f_k(x) \approx \text{Normal}(x;\ \mu_k,\ v_k)$$
$$v_k = \tau_k^2 + w\, \mu_k^2 + s_0^2.$$

Approximation: the implementation uses $w\, \mu_k^2$. It does not perform the exact heteroscedastic marginalization over every physical duration, which would involve $E[s^2]$. The real stimulus distributions are truncated and non-overlapping, whereas the Gaussian approximations have support over the whole real line.

3. Bayesian log odds and decision regions

Given a pre-choice long-category prior π, the cost-adjusted posterior log odds are

$$\Lambda(x) = \log[f_1(x) / f_0(x)] + \operatorname{logit}(\pi) - \log(\kappa)$$
$$= a_2x^2 + a_1x + a_{0,t}$$

$$a_2 = 1/(2v_0) - 1/(2v_1)$$
$$a_1 = \mu_1/v_1 - \mu_0/v_0$$
$$a_{0,t} = \mu_0^2/(2v_0) - \mu_1^2/(2v_1)$$
$$+ 0.5 \log(v_0/v_1) + \operatorname{logit}(\pi) - \log(\kappa).$$

The non-lapse response is long wherever Λ(X) > 0. The positive parameter κ is a decision-cost ratio: κ=1 is neutral, κ>1 favors short, and κ<1 favors long.

M0 decision convention

M0 fixes π=0.5 and requires one positive boundary T satisfying Λ(T)=0. Its registered action rule is

$$\text{long iff } X > T$$
$$q = P(X > T \mid s) = \Phi[(s - T)/\sigma].$$

M0 deliberately discards any mathematically valid lower tail below a negative quadratic root. Unsupported root structures raise an error.

M1 through M4 decision convention

The dynamic models integrate the complete region where the quadratic is positive. For two ordered roots $r_- < r_+$, define F(x)=Φ[(x-s)/σ]. Then

$$\text{if } a_2 > 0\text{: } q = F(r_-) + 1 - F(r_+) \text{ [outside the roots]}$$
$$\text{if } a_2 < 0\text{: } q = F(r_+) - F(r_-) \text{ [between the roots]}.$$

Linear, double-root, no-real-root, and constant-sign cases are handled explicitly. Therefore M0 and M1-M4 do not have globally identical decision-region definitions.

4. Category-dependent lapse mixture

The final observable choice probability is obtained by mixing q with a random-choice lapse:

$$\varepsilon = \square_{\text{long}}\ I(C=1) + \square_{\text{short}}\ I(C=0)$$
$$p = (1 - \varepsilon)\ q + \varepsilon/2.$$

Conditional on a lapse, the choice is uniform over the two alternatives. A lapse rate of one therefore produces p=0.5.

Causal qualification: current category feedback cannot affect the current Bayesian prior π or non-lapse probability q. However, the current true category does select ℓ_{short} versus ℓ_{long}, so it can affect the final current probability p and its likelihood.

5. M0: sensory-only model

$$\pi = 0.5 \text{ for every trial.}$$

M0 has no block state, trial-history dependence, or learning state. Apart from category-dependent lapse, two trials with the same physical duration have the same prediction.

Fitted vector: $\Theta_{M0} = (w, s_0, \ell_{short}, \ell_{long}, \kappa)$.

6. M1: constant-hazard HMM with fixed prototypes

M1 introduces a latent block Z in state order (N,S,L). The fixed probability of a long-category stimulus in each state is

$$\theta = (\theta_N, \theta_S, \theta_L) = (0.5, 0.2, 0.8).$$

Let $\xi^-(b)=P(Z=b \mid \text{earlier valid feedback})$ be the pre-choice block belief. The category prior entering the sensory decision is

$$\pi = \Sigma_b \, \xi^-(b) \, \theta_b.$$

After the current choice has been predicted, valid category feedback C updates the belief:

$$e_b(C) = \theta_b^{C} \, (1-\theta_b)^{1-C}$$
$$\xi^+(b) = \xi^-(b) e_b(C) \, / \, \Sigma_j \, \xi^-(j) e_j(C).$$

The biased-state hazard is $h_B=\text{sigmoid}(a_h)$. Neutral uses the fixed tie $h_N=h_B/2$. With first-biased-block probability $\omega=0.5$:

$$T = \begin{bmatrix} 1-h_N & h_N(1-\omega) & h_N\omega \\ 0 & 1-h_B & h_B \\ 0 & h_B & 1-h_B \end{bmatrix}$$
$$\xi^- = \xi^+ T.$$

Neutral cannot be re-entered, and short- and long-biased blocks alternate. The geometric mean durations are $E[D_B]=1/h_B$ and $E[D_N]=2/h_B$. Every session resets to $\xi_1^-=(1,0,0)$.

Fitted vector: $\Theta_{M1} = (\Theta_{M0}, a_h)$.

7. M2: fitted static subjective prototypes

M2 keeps M1's constant-hazard HMM but estimates how strongly the animal expects the biased blocks to differ:

$$\theta_N = 0.5$$
$$\theta_S = \text{sigmoid}(-\delta_0)$$
$$\theta_L = \text{sigmoid}(\delta_0) = 1 - \theta_S.$$

- Small δ_0: weak subjective difference between biased blocks.
- Large δ_0: strong subjective difference between biased blocks.

The same prototype vector is used in both the choice prior and the feedback emission update.

Exact M1 restriction: at $\delta_0=\log(4)$, the prototypes are exactly (0.5,0.2,0.8), so M2 reproduces M1 trial by trial.

At δ_0=0, all prototypes are 0.5. The prior remains 0.5 regardless of belief, making the hazard choice-unidentifiable. This gives M0-like history independence, although M2 continues to use the full quadratic decision-region solver.

Fitted vector: $\Theta_{M2} = (\Theta_{M1}, \delta_0)$.

From static belief to adaptive inference

M2 - static contrast

One fitted delta0 defines the short- and long-block prototypes for every trial and every analyzed session.

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delta_t = delta0
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M3 - learned contrast

Feedback changes delta within a session using posterior block responsibilities. Every session still starts again at delta0.

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delta_(t+1) = max(10^-6, delta_t + alpha_theta u_t)
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M4 - learned contrast plus block age

The same learning rule is retained, while the switch hazard becomes a function of inferred run length.

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h_B(r) = sigmoid[a_h + b_h(r - 23)]
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8. M3: responsibility-weighted prototype learning

M3 allows the subjective contrast to change within a session. It resets at the start of every session:

$$\delta_1 = \delta_0$$
$$\theta = (0.5, \text{sigmoid}(-\delta), \text{sigmoid}(\delta)).$$

After feedback, let $\gamma_{b,t}=\xi^{+}(b)$ be the posterior responsibility assigned to block b. The local learning signal is

$$u = \gamma_{L,t}(C-\theta_{L,t}) - \gamma_{S,t}(C-\theta_{S,t}).$$

The contrast update is

$$\delta^{*} = \delta + \alpha_{\theta}u$$
$$\delta = \max(10^{-6}, \delta^{*}).$$

The neutral responsibility has no direct learning term. Evidence consistent with the believed biased state tends to increase contrast; inconsistent evidence tends to decrease it. The hard floor keeps the short and long prototypes ordered, but it creates a non-smooth boundary.

Exact M2 restriction: $\alpha_{\theta}=0$ freezes $\delta=\delta_0$, so M3 reproduces M2 exactly.

Fitted vector: $\Theta_{M3} = (\Theta_{M2}, \alpha_{\theta})$.

9. M4: duration-aware capped-tail HSMM

M4 tracks both block identity and inferred run length:

$$\xi^{-}(b,r) = P(Z=b, R=r \mid \text{earlier feedback})$$
$$b \in \{N,S,L\},\ r \in \{1,\ldots,65\}.$$

The final state r=65 means R≥65. It is a capped tail, not a forced switch and not a claim that the block is exactly 65 trials long.

The biased-block hazard depends on run length:

$$h_B(r) = \text{sigmoid}[a_h + b_h(r-23)]$$
$$h_N(r) = h_B(r)/2.$$

- a_h is the hazard logit specifically at run length 23.
- $b_h\geq 0$ imposes a nondecreasing switch hazard as the block ages.

After feedback conditioning, write $q_b(r)=\xi^{+}(b,r)$. Each source mass is divided into

$$\text{survival mass} = q_b(r)[1-h_b(r)]$$
$$\text{switch mass} = q_b(r)h_b(r).$$

Survival moves to (b,r+1), while survival from r=65 remains in the tail. Every switch resets run length to one. Short switches to long, long switches to short, and neutral exit is divided between short and long using ω=0.5.

The block marginal and current choice prior are

$$\xi^{-}(b) = \Sigma_r\ \xi^{-}(b,r)$$
$$\pi = \Sigma_b\ \xi^{-}(b)\ \theta_{b,t}.$$

Exact M3 restriction: when $b_h=0$, $h_B(r)=\text{sigmoid}(a_h)$ for every r. Marginalizing over run length then reproduces M3 exactly, including beyond the 65-trial cap.

Fitted vector: $\Theta_{M4} = (\Theta_{M3}, b_h)$.

10. Causal order, omissions, and session resets

Step	M1-M4 operation	Can affect current choice?
1	Start from pre-choice latent belief.	Yes
2	Construct current prototypes.	Yes
3	Compute π from the pre-choice belief.	Yes
4	Compute the Bayesian non-lapse probability q.	Yes
5	Select category-specific lapse and compute p.	Yes
6	Score a valid observed choice.	Likelihood only
7	Condition latent belief on valid category feedback.	Future trials
8	M3/M4: update δ.	Future trials
9	Apply HMM or HSMM transition.	Future trials

- An omitted or invalid choice contributes no Bernoulli likelihood term.
- If valid category feedback exists on that trial, belief and learning can still update.
- The latent transition advances even across an omission, except that no transition is needed after the final trial.
- Belief, run length, and δ reset independently at every session boundary.

11. Likelihood and optimization target

For valid choices, the observation model is Bernoulli:

$$Y_{dt} \sim \text{Bernoulli}(p_{dt}).$$

For the valid-decision set V_d in session d:

$$J_d = -\Sigma_{t \in V_d} [Y_{dt} \log p_{dt} + (1-Y_{dt}) \log(1-p_{dt})]$$
$$N_d = |V_d|.$$

The primary pooled-trial objective is

$$J_{pooled} = (\Sigma_d J_d) / (\Sigma_d N_d).$$

The supported equal-session sensitivity objective is

$$J_{equal} = (1/D) \Sigma_d J_d/N_d.$$

The two objectives differ when session lengths differ. The registered primary objective weights every valid trial equally.

Choice-only fitting: category-emission probabilities are used inside the latent-state filtering and learning recursion, but log P(C|Z) is not added to the outer objective. M1-M4 are therefore generative latent-state filters fitted with a conditional choice likelihood, not full joint likelihood models of categories and choices.

12. Parameter constraints

Parameter	Models	Constraint / transform	Scientific meaning
w	M0-M4	softplus(raw)+ε	Duration-dependent variance coefficient.
s_0	M0-M4	softplus(raw)+ε	Baseline sensory SD in seconds.
ℓ_{short}, ℓ_{long}	M0-M4	sigmoid(raw)	Category-specific lapse probabilities.
κ	M0-M4	exp(raw)	Positive decision-cost ratio.
a_h	M1-M4	unconstrained; h=sigmoid(a_h)	Hazard logit.
δ_0	M2-M4	softplus(raw)+ε	Initial/static subjective prototype contrast.
α_θ	M3-M4	sigmoid(raw)	Prototype learning step size.
b_h	M4	softplus(raw)+ε	Nonnegative run-length hazard slope.

13. Exact restrictions and hierarchy

$$M4 \text{ -- } b_h=0 \text{ --> } M3 \text{ -- } \alpha_\theta=0 \text{ --> } M2 \text{ -- } \delta_0=\log(4) \text{ --> } M1$$

M2 at $\delta_0=0$ gives a fixed 0.5 prior and removes behavioral history dependence, but M0 remains a separate readout branch because M0 uses the positive-upper-threshold convention while M1-M4 integrate the complete quadratic sign region.

14. Interpretation and identifiability cautions

- Sensory noise versus lapse: w, s_0, and the lapse rates can all flatten psychometric curves and may be difficult to separate without informative extreme-duration trials.
- Prior versus choice bias: only logit(π) - log(κ) enters the log-odds intercept. A constant prior displacement is confounded with κ; trial-varying priors provide the main source of separation.
- Hazard versus prototype contrast: h_B and δ_0 both affect the size and persistence of history-dependent priors, producing a potential ridge in M2 and later models.
- Category-specific lapses: the two lapse rates can absorb category asymmetries that could otherwise be attributed to noise, bias, or latent-state inference.
- M3 learning: a positive α_θ alone does not establish meaningful learning. Responsibilities, actual update signals, held-out improvement, and projection frequency must also be examined.
- M4 hazard shape: $b_h \geq 0$ allows only constant or increasing hazard. Decreasing and non-monotonic duration dependence are outside this model family.
- M4 tail: once mass reaches $R \geq 65$, the model uses h(65). It cannot distinguish hazards at run lengths 66, 80, or 100. Large tail occupancy weakens interpretation of b_h.
- Conditional evidence: dynamic parameters are supported only insofar as category-driven latent trajectories improve future choice prediction. They are not fitted directly to reproduce the category sequence.
- Fixed task moments: uncertainty in μ_k and τ_k^2 is not propagated into parameter uncertainty.

15. Predictive metrics used in the repository

$$\text{ELPD} = \sum_i [Y_i \log p_i + (1-Y_i) \log(1-p_i)]$$
$$\text{Mean NLL} = -\text{ELPD} / N$$
$$\text{Brier score} = (1/N) \sum_i (Y_i - p_i)^2$$
$$\text{Accuracy} = (1/N) \sum_i I[(p_i > 0.5) = Y_i].$$

Higher ELPD and lower mean NLL or Brier score indicate better probabilistic predictions. Threshold accuracy discards probability calibration information and is therefore less informative than likelihood for comparing these models.

Sliding-window and single-session analyses reuse the same model equations and likelihood on different chronological subsets. They are analysis strategies, not additional generative model families.

16. One-sentence interpretation of each model

Model	Interpretation
M0	Choices arise from noisy duration measurement, bias, and category-specific lapse, with no learned task context.
M1	The animal performs Bayesian inference over known neutral/short-biased/long-biased block statistics.
M2	The animal performs the same inference but has its own fitted belief about the strength of block bias.
M3	The animal updates that subjective block contrast from category feedback within each session.
M4	The animal additionally expects block-switch probability to depend on inferred block age.

End of mathematical explanation